Random Variables

1.1. Introduction

A random variable associates each outcome of a random experiment with a probability and it represents a measurable aspect or property of the outcomes. For these reasons, a random variable is essential to be defined to assume a real number to each and every outcome of the random experiment.

1.2. Random Variable

Let S be a sample space of a random experiment. A random variable is a function that associates a real number with each element in the sample space, S. In other words, we say that a random variable is a real valued function defined on the sample space, S. That is, the function $X(s) = x$ that maps the elements of the sample space S into real numbers is called the random variable associated with the concerned experiment.

Example 1: In the experiment of throwing a coin twice, the sample space, S = {HH, HT, TH, TT}. Define a function X from S into R, a set of all real numbers as $X(s) = x$ (the number of heads). That is, $X(HH) = 2$; $X(HT) = 1$; $X(TH) = 1$ and $X(TT) = 0$. Thus, X is a random variable defined on S.

Example 2: In a cinema theatre, the number of people arriving at any t , X in the interval from 1:30 pm to 2:30 pm is a real number. Therefore, X is a random variable.

1.3. Types of Random Variables

Random variables are commonly classified into two types namely, discrete random variables and continuous random variables.

1.4. Discrete Random Variable

A random variable on a random sample S is said to be discrete if the range of X is countable. In other words, a discrete random variable is a random variable that can assume at most a finite or a countably infinite number of possible values.

Some examples of the discrete random variables are:

1. The number heads appearing in tossing ten coins at a time.
2. The number defective article produced in the morning shift in the factory.
3. The number of students absent in the French class.

1.5. Probability Function (or) Probability Mass Function of a Discrete Random Variable

Let X be a discrete random variable on a sample space S, which can take the values $x_1, x_2, \dots x_n$ such that $P(X = x_i) = P(x_i) = p_i$ (called the probability of x_i). The function p_i is called the probability mass function (p.m.f) of the random variable X if p_i must satisfy the following conditions: (i) $p_i \geq 0$ for all i and (ii) $\sum_i^n p_i = 1$

The table containing the values of the random variable, $X = x_i, i = 1,2,\dots n$ and its probabilities, $P(X = x_i) = p_i$, $i = 1,2,\dots n$ is called the probability distribution of the random variable X. That is,

$X = x_i$	x_1	$\cdots$	x_i	$\cdots$	x_n
$P(X = x_i) = p_i$	p_1	...	p_i	...	p_n

1.6. Distribution Function of the Discrete Random Variable

The distribution function or briefly the cumulative distribution function (c.d.f) for a discrete random variable X is defined by $F_X(x) = F(x) = P(X \leq x) = \sum_{-\infty}^{x} P(X = x_i)$ where x is any real number, that is, $-\infty < x < \infty$.

The distribution function $F(x)$ has the following properties:

(i) $F(x)$ is non-decreasing $[F(x) \leq F(y)$ if $x \leq y]$

(ii) $\lim_{x \to -\infty} F(x) = 0$; $\lim_{x \to \infty} F(x) = 1$

(iii) $F(x)$ is continuous from the right $\left[\text{that is, } \lim_{h \to 0^+} F(x+h) = F(x) \text{ for all } x\right]$

1.7. Mean and Variance of a Discrete Random Variable

The mean value of a discrete random variable X is defined as $\overline{X} = \sum_{i=1}^{n} x_i p_i$ and the variance of a discrete random variable X is defined as

$$Var(X) = \sum_{i=1}^{n} \left((x_i - \overline{X})^2 p_i\right) = \sum_{i=1}^{n} x_i^2 p_i - \overline{X}^2.$$

Example 4: Let the random variable X denote the number of heads in three tosses of a fair coin. Find the probability distribution of X. Also obtain the cumulative probability distribution.

Solution: Let X denote the number of heads in three tosses of a fair coin.
$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
Here $n(s) = 8$ and the range of $X = \{0,1,2,3\}$.

$$P(X = 0) = P \text{ (no head)} = \frac{1}{8}; \quad P(X = 1) = P \text{ (one head)} = \frac{3}{8}$$

$$P(X = 2) = P \text{ (two heads)} = \frac{3}{8}; \quad P(X = 3) = P \text{ (three heads)} = \frac{1}{8}$$

Thus the probability distribution of X is

$X = x$	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Now, the cumulative probability of X, $F(x) = P(X \le x)$.

Now, $F(0) = P(X = 0) = \frac{1}{8}$

$$F(1) = P(X = 0) + P(X = 1) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$$

$$F(2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$$

$$F(3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1.$$

Therefore, cumulative probability distribution

$F(x) = 0$ when $x < 0$; $F(x) = \frac{1}{8}$ when $0 \le x < 1$; $F(x) = \frac{4}{8}$ when $1 \le x < 2$;

$F(x) = \frac{7}{8}$ when $2 \le x < 3$; $F(x) = 1$ when $x \ge 3$.

Example 5: A pair of dice is rolled once. Let X be the random variable whose value for any outcome is the sum of the two numbers on the dice.

(i) Find the probability distribution of X.
(ii) Find the probability that X is an odd number.
(iii) Find $P(5 \le X \le 8)$.

Solution: Let X be the random variable whose values for any outcome is the sum of the two numbers on the dice.

(i) The sample space, $S = \begin{bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{bmatrix}$

Here $n(S) = 36$ and the range of $X = \{2,3,4,5,6,7,8,9,10,11,12\}$.

Therefore, the probability distribution of X is

$X = x$	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(ii) P(X is an odd number) $= P(X = 3) + P(X = 5) + P(X = 7) + P(X = 9) + P(X = 11)$

$$= \frac{2}{36} + \frac{4}{36} + \frac{6}{36} + \frac{4}{36} + \frac{2}{36} = \frac{1}{2}$$

(iii) $P(5 \le X \le 8) = P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8)$

$$= \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} = \frac{20}{36}$$

Example 6: A random variable X has the following probability function:

Values of $X = x$	0	1	2	3	4
Probability $P(x)$	$3a$	$4a$	$6a$	$7a$	$8a$

(i) Determine the value of a.

(ii) Find $P(X \le 2)$, $P(X > 3)$ and $P(0 < X < 4)$.

Solution: (i) We know that if $P(x)$ is the probability mass function, then $\sum_x P(x) = 1$.

Therefore, $3a + 4a + 6a + 7a + 8a = 1$

That is, $28a = 1$

This implies that $a = \frac{1}{28}$

Therefore, the probability distribution of X is

Values of X	0	1	2	3	4

Probability $P(x)$	$\frac{3}{28}$	$\frac{4}{28}$	$\frac{6}{28}$	$\frac{7}{28}$	$\frac{8}{28}$

(ii) Now, $P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2) \; = \frac{3}{28} + \frac{4}{28} + \frac{6}{28} \quad = \frac{13}{28}$.

Now, $P(X > 3) = P(X = 4) = \frac{8}{28}$.

Now, $P(0 < X < 4) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{4}{28} + \frac{6}{28} + \frac{7}{28} \quad = \frac{17}{28}$.

Example 7: A random variable X has the following probability function:

Values of X	-2	-1	0	1
Probability $P(x)$	0.4	k	0.2	0.3

Find (a) the value of k (b) $P(X < 1)$ (c) the distribution function of X .

Solution: (a) We know that if $P(x)$ is the probability mass function, then $\sum_x P(x) = 1$.

Therefore, $0.4 + k + 0.2 + 0.3 = 1$
That is, $0.9 + k = 1$
This implies, $k = 0.1$
Therefore, the probability distribution of X is

Values of X	-2	-1	0	1
Probability $P(x)$	0.4	0.1	0.2	0.3

(b) Now, $P(X < 1) = P(X = -2) + P(X = -1) + P(X = 0) = 0.4 + 0.1 + 0.2 \; = 0.7$

(c) Now, the cumulative probability of X , $F(x) = P(X \le x)$.

Now, $F(-2) = P(X \le -2) = 0.4$

$F(-1) = P(X \le -1) = P(X = -2) + P(X = -1) = 0.5$

$F(0) = P(X \le 0) = P(X = -2) + P(X = -1) + P(X = 0) = 0.7$

$F(1) = P(X \le 1) = P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1) = 1.$

Therefore, cumulative probability distribution is given below:
$F(x) = 0$ when $x < -2$; $F(x) = 0.4$ when $-2 \le x < -1$; $F(x) = 0.5$ when $-1 \le x < 0$;
$F(x) = 0.7$ when $0 \le x < 1$; $F(x) = 1$ when $x \ge 1$.

Example 8: The monthly demand for Titan watches is known to have the following probability function:

Demand	1	2	3	4	5	6	7	8

Probability	0.08	0.12	0.19	0.24	0.16	0.10	0.07	0.04

Determine the expected demand for watches and also compute the variance.

Solution: Expected demand for watches $\overline{X} = \sum_{i=1}^{n} x_i p_i$

$$= 1\times 0.08 + 2\times 0.12 + 3\times 0.19 + 4\times 0.24 + 5\times 0.16 + 6\times 0.10 + 7\times 0.07 + 8\times 0.04$$

$$= 4.06$$

Now, $\sum_{i=1}^{n} x_i^2 p_i = 1\times 0.08 + 4\times 0.12 + 9\times 0.19 + 16\times 0.24$

$$+ 25\times 0.16 + 36\times 0.10 + 49\times 0.07 + 64\times 0.04 = 19.7$$

Now, Variance, $Var(X) = \sum_{i=1}^{n} \left((x_i - \overline{X})^2 p_i\right) = \sum_{i=1}^{n} x_i^2 p_i - \overline{X}^2 = 19.7 - (4.06)^2 = 3.2164$.

Example 9: If the random variable takes the values 1,2,3 and 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$, find the probability distribution and the cumulative distribution function of X.

Solution: Let $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k$

That is, $P(X=1) = \frac{k}{2}$, $P(X=2) = \frac{k}{3}$, $P(X=3) = k$, $P(X=4) = \frac{k}{5}$

We know that if $P(x)$ is the probability mass function, then $\sum_{x} P(x) = 1$.

Therefore, $\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$.

This implies that, $k = \frac{30}{61}$.

The probability distribution of X is given by

x_i	1	2	3	4
$P(x_i)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

Now, the cumulative distribution function, $F(x) = P(X \le x)$.

Now, $F(1) = P(X \le 1) = \frac{15}{61}$

$$F(2) = P(X \le 2) = P(X=1) + P(X=2) = \frac{25}{61}$$

$F(3) = P(X \le 3) = P(X = 1) + P(X = 2) + P(X = 3) = \dfrac{55}{61}$

$F(4) = P(X \le 4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$.

Therefore, the cumulative distribution function

$F(x) = 0$ when $x < 1$; $F(x) = \dfrac{15}{61}$ when $1 \le x < 2$; $F(x) = \dfrac{25}{61}$ when $2 \le x < 3$;

$F(x) = \dfrac{55}{61}$ when $3 \le x < 4$; $F(x) = 1$ when $x \ge 4$.

Exercise 1.1

1. A random variable X has the following probability function

Values of X	1	3	4	7	9	10	14	18
Probability $P(x)$	0.11	0.07	0.13	0.28	0.18	0.05	0.12	0.06

Find $P(X \ge 10)$, $P(X < 4)$, $P(4 \le X \le 9)$.

Solution: $P(X \ge 10) = 0.23$**;** $P(X < 4) = 0.18$**;** $P(4 \le X \le 9) = 0.59$

2. A random variable X has the following probability function.

Values of X	0	1	2	3	4	5	6	7	8
$P(X = x)$	c	$3c$	$5c$	$7c$	$9c$	$11c$	$13c$	$15c$	$17c$

(i) Find the value of c (ii) Find $P(X < 3)$, $P(0 < X < 3)$, $P(X \ge 3)$

(iii) Find the distribution function of X.

Solution: (i) $c = \dfrac{1}{81}$, (ii) $P(X < 3) = \dfrac{1}{9}$, $P(0 < X < 3) = \dfrac{8}{81}$, $P(X \ge 3) = \dfrac{8}{9}$

(iii) Distribution function of X

$F(x) = 0$ when $x < 0$; $F(x) = \dfrac{1}{81}$ when $0 \le x < 1$; $F(x) = \dfrac{4}{81}$ when $1 \le x < 2$;

$F(x) = \dfrac{9}{81}$ when $2 \le x < 3$; $F(x) = \dfrac{16}{81}$ when $3 \le x < 4$; $F(x) = \dfrac{25}{81}$ when $4 \le x < 5$

$F(x) = \dfrac{36}{81}$ when $5 \le x < 6$; $F(x) = \dfrac{49}{81}$ when $6 \le x < 7$; $F(x) = \dfrac{64}{81}$ when $7 \le x < 8$

$F(x) = 1$ when $x \ge 8$.

3. A discrete random variable X has the probability function given below:

Values of X	0	1	2	3	4	5	6	7
$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find (i) the value of k , (ii) $P(X < 6)$, $P(0 < X < 4)$, $P(X \geq 6)$

(iii) the distribution function of X .

Solution: (i) $a = 0.10$ (or) -1,

(ii) $P(X < 6) = 0.81$, $P(X \geq 6) = 0.19$, $P(0 < X < 4) = 0.5$

(iii) Distribution function of X

$F(x) = 0$ when $x < 0$; $F(x) = 0$ when $0 \leq x < 1$; $F(x) = 0.10$ when $1 \leq x < 2$;

$F(x) = 0.30$ when $2 \leq x < 3$; $F(x) = 0.50$ when $3 \leq x < 4$; $F(x) = 0.80$ when $4 \leq x < 5$

$F(x) = 0.81$ when $5 \leq x < 6$; $F(x) = 0.83$ when $6 \leq x < 7$; $F(x) = 1$ when $x \geq 7$.

4. If the probability mass function of a random variable X is given by $P(X = r) = kr^3$; $r = 1, 2, 3, 4$. Find (i) the value of k (ii) the probability distribution.

Solution: (i) $k = 0.01$

(ii) Probability distribution

x	1	2	3	4
$P(x)$	0.01	0.08	0.27	0.64

5. The number of messages sent per hour over a computer network has the following distribution

Number of messages(x)	10	11	12	13	14	15
Probability $P(x)$	0.08	0.15	0.30	0.20	0.20	0.07

Determine the mean and standard deviation of the number of messages.

Solution: Mean $= 12.5$, S.D$(X) = 1.36$.

6. A random variable X has the following probability distribution

Values of X	-2	-1	0	1	2	3
$P(X = x)$	0.1	K	0.2	2K	0.3	3K

(i) Find K, (ii) Evaluate $P(X < 2)$ and $P(-2 < X < 2)$ (iii) find the c.d.f. of X .

Solution: (i) $K = 0.07$ (ii) $P(X < 2) = 0.50$ and $P(-2 < X < 2) = 0.40$

(iii) c.d.f of X is given below:

$F(x) = 0$ when $x < -2$; $F(x) = 0.10$ when $-2 \leq x < -1$; $F(x) = 0.17$ when $-1 \leq x < 0$

$F(x) = 0.37$ when $0 \leq x < 1$; $F(x) = 0.50$ when $1 \leq x < 2$; $F(x) = 0.80$ when $2 \leq x < 3$;

$F(x) = 1$ when $x \geq 3$..

2.1 Continuous random variable

A random variable X is said to be continuous if it takes all possible values between certain limits or in an interval which may be finite or infinite. In other words, if a random variable can assign any value in a given range of values, then such a random variable is known as the continuous random variable.

Some examples of the continuous random variables are:

4. The computer time (in seconds) required to process a certain program.
5. The amount of rain falls in the certain city.
6. The heat gained by a ceiling fan when it has worked for one hour.

2.2. Probability Density function of a Continuous Random Variable

If X is a continuous random variable such that $P\{x-\frac{dx}{2} \le X \le x+\frac{dx}{2}\} = f(x)dx$, then $f(x)$ is called the probability density function (p.d.f) of X provided $f(x)$ satisfies the following conditions:

(i) $f(x) \ge 0$, for all $x \in R$; (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ and (iii) $P(a \le x \le b) = \int_a^b f(x)dx$.

2.3. Distribution Function of the Continuous Random Variable

The cumulative distribution function (c.d.f.) or briefly the distribution function for a continuous random variable X is defined by $F_X(x) = F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)dx$, $-\infty < x < \infty$.

The distribution function $F(x)$ has the following properties:

(i) $0 \le F(x) \le 1$, $-\infty < x < \infty$.

(ii) $\lim_{x\to-\infty} F(x) = 0$; $\lim_{x\to\infty} F(x) = 1$

(iii) $F(x)$ is continuous function of x on the right.

(iv) $P(a \le x \le b) = \int_a^b f(x)dx = F(b) - F(a)$.

2.4. Relationship between probability distribution function $F(x)$ and probability density function $f(x)$:

From the definition of probability distribution function, we can conclude that $\frac{d}{dx}(F(x)) = f(x)$**,** that is, the probability density function of a continuous random variable X , $f(x)$ is the derivative of the probability distribution function of the random variable, $F(x)$

1.8. Mean and Variance of a Continuous Random Variable

The mean value of a continuous random variable X is defined as $\overline{X} = \int_{-\infty}^{\infty} xf(x)dx$ and the variance of a continuous random variable X is defined as $Var(X) = \int_{-\infty}^{\infty} (x-\overline{X})^2 f(x)dx$

Example 1: Diameter of an electric cable, X is assumed to be a continuous random variable with the function

$$f(x) = \begin{cases} 6x(1-x) & ;\text{for} \quad 0 \le x \le 1 \\ 0 & ;\text{for} \quad \text{otherwise} \end{cases}$$

Check whether $f(x)$ is a probability density function or not.

Solution: From property (i) for p.d.f, it is obvious that for $0 \le x \le 1$, $f(x) \ge 0$.
Now, for property (ii) consider,

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^1 6x(1-x)dx = \int_0^1 6(x-x^2)dx = 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 1$$

Therefore, $f(x) = 6x(1-x); \quad 0 \le x \le 1$ is a p.d.f of a random variable X.

Example 2: Is the function defined as follows a density function? $f(x) = \begin{cases} 2e^{-x} & ,x \ge 0 \\ 0 & ,x < 0 \end{cases}$

Solution: From property (i) for p.d.f, it is obvious that for $0 \le x \le 1$, $f(x) \ge 0$.

Now, for property (ii) consider, $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} f(x)dx + \int_0^{\infty} f(x)dx$

$$= \int_{-\infty}^{0} 0dx + 2\int_0^{\infty} e^{-x}dx$$

$$= 0 + 2[-e^{-x}]_0^{\infty} = 2(-e^{-\infty}+1) = 2$$

This implies that, $f(x)$ does not satisfy the condition of the density function.
Therefore, the given function is not a p.d.f. of a random variable.

Example 3: A random variable X has the p.d.f $f(x) = cx^2(1-x), 0 < x < 1$. Find (i) the constant c (ii) $P(X < \frac{1}{2})$.

Solution: (i) We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$

That is, $\int_0^1 cx^2(1-x)dx = 1$. This implies that, $c = \frac{1}{12}$.

(ii) $P(X < \frac{1}{2}) = \frac{1}{12}\int_0^{\frac{1}{2}} x^2(1-x)dx = \frac{5}{16}$

Example 4: A continuous random variable X can assume any value between $x = 2$ and $x = 5$ has a density function given by $f(x) = k(1+x)$. Find $P(X < 4)$.

Solution: We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$

That is, $\int_2^5 k(1+x)dx = 1$

This implies that, $k\left[\frac{(1+x)^2}{2}\right]_2^5 = 1$

This implies that, $k = \frac{2}{27}$

Therefore, $P(X < 4) = P(2 < X < 4) = \int_2^4 k(1+x)dx = \frac{16}{17}$.

Example 5: The length of time (in minutes) that a certain lady speaks on the telephone is found to be random phenomenon, with a probability function specified by the probability density function $f(x)$ as

$$f(x) = \begin{cases} Ae^{-\frac{x}{5}} & ; x \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

(i) Find the value of *A*.
(ii) What is the probability that the number of minutes that she will talk over the phone is (a) more than 10 minutes (b) less than 5 minutes (c) between 5 and 10 minutes.

Solution: (i) We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$

Therefore, $\int_0^{\infty} Ae^{-\frac{x}{5}}dx = 1$

That is, $A\left[\frac{e^{-\frac{x}{5}}}{\frac{-1}{5}}\right]_0^{\infty} = 1$. This implies that $A = \frac{1}{5}$

(ii) Let X be the length of time that the lady speaks

(a) P(more than 10 minutes) = $P(X > 10) = \int_{10}^{\infty} \frac{1}{5} e^{-\frac{x}{5}} dx = \frac{1}{5}\left[\frac{e^{-\frac{x}{5}}}{\frac{-1}{5}}\right]_{10}^{\infty} = e^{-2}$

(b) P(less than 5 minutes) = $P(X < 5) = \int_{0}^{5} \frac{1}{5} e^{-\frac{x}{5}} dx = \frac{1}{5}\left[\frac{e^{-\frac{x}{5}}}{\frac{-1}{5}}\right]_{0}^{5} = [1 - e^{-1}]$

(c) P(between 5 and 10 minutes) = $P(5 < X < 10) = \int_{5}^{10} \frac{1}{5} e^{-\frac{x}{5}} dx = \frac{1}{5}\left[\frac{e^{-\frac{x}{5}}}{\frac{-1}{5}}\right]_{5}^{10}$

$$= [e^{-1} - e^{-2}]$$

Example 6: If a random variable X has the p.d.f $f(x) = \begin{cases} \frac{1}{2}(x+1); & -1 < x < 1 \\ 0 \quad ; & \text{otherwise} \end{cases}$.

Find the mean and variance of X.

Solution: Mean = $\int_{-1}^{1} x f(x) dx = \frac{1}{2}\int_{-1}^{1} x(x+1) dx = \frac{1}{3}$

Variance = $\int_{-1}^{1} (x - mean)^2 f(x) dx = \int_{-1}^{1} (x - \frac{1}{3})^2 \frac{(x+1)}{2} dx = \frac{1}{18}\int_{-1}^{1} (9x^3 + 3x^2 - 5x + 1) dx$

$$= \frac{2}{9}$$

Example 7: A continuous random variable X has a p.d.f. $f(x) = 3x^2, 0 \le x \le 1$, Find 'a' such that $P(X \le a) = P(X > a)$.

Solution: Given that $P(X \le a) = P(X > a)$

Since the total probability is 1, we have $P(X \le a) = \frac{1}{2}$ and $P(X > a) = \frac{1}{2}$

Now, since $P(X \le a) = \frac{1}{2}$, we have $\int_{0}^{a} f(x) dx = \frac{1}{2}$.

That is, $\int_0^a 3x^2 dx = \frac{1}{2}$. This implies $a = \left(\frac{1}{2}\right)^{\frac{1}{3}}$

Example 8: If the density function of a continuous random variable X is given by

$$f(x) = \begin{cases} ax & ,0 \le x \le 1 \\ a & ,1 \le x \le 2 \\ 3a - ax & ,2 \le x \le 3 \\ 0 & ,\text{otherwise} \end{cases}$$

(i) Find the value of a (ii) Find the c.d.f of X .

Solution: (i) We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$

Now, $\int_0^3 f(x)dx = 1$

This implies $\int_0^1 axdx + \int_1^2 adx + \int_2^3 (3a - ax)dx = 1$

That is, $a = \frac{1}{2}$

(ii) Now, the cumulative distribution function, $F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)dx$

When 'x' lies in $0 \le x \le 1$,

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{-\infty}^{0} f(x)dx + \int_0^x f(x)dx = 0 + \int_0^x axdx = a\left(\frac{x^2}{2}\right).$$

When 'x' lies in $1 \le x \le 2$,

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{-\infty}^{0} f(x)dx + \int_0^1 f(x)dx + \int_1^x f(x)dx = 0 + \int_0^1 axdx + \int_1^x adx$$

$$= \frac{a}{2} + ax - a .$$

When 'x' lies in $2 \le x \le 3$,

$$F(x) = \int_{-\infty}^{0} f(x)dx + \int_0^1 f(x)dx + \int_1^2 f(x)dx + \int_2^x f(x)dx$$

$$= 0 + \int_0^1 axdx + \int_1^2 adx + \int_2^x (3a - ax)dx = 3ax - \frac{ax^2}{2} - \frac{5a}{2}$$

Example 9: A random variable X has the density function $f(x) = \begin{cases} \dfrac{K}{1+x^2} & ,-\infty < x < \infty \\ 0 & ,otherwise \end{cases}$

Find 'K' and the distribution function $F(x)$.

Solution: We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$.

Now, $K\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx = 1$.

This implies, $K[\tan^{-1} x]_{-\infty}^{\infty} = 1$.

That is , $K\left(\frac{\pi}{2}+\frac{\pi}{2}\right) = 1$. This implies, $K = \frac{1}{\pi}$

Now, the cumulative distribution function, $F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)dx$

$$= \frac{1}{\pi}\int_{-\infty}^{x} \frac{1}{1+x^2}dx \quad = \frac{1}{\pi}[\tan^{-1} x]_{-\infty}^{x} = \frac{1}{\pi}[\tan^{-1} x + \frac{\pi}{2}].$$

Example 10: For the following density function $f(x) = a\,e^{-|x|}, -\infty < x < \infty$, find (i) the value of 'a' (ii) mean and variance.

Solution:(i) We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$.

Now, $\int_{-\infty}^{\infty} a\,e^{-|x|} = 1$.

Since $e^{-|x|}$ is an even function , we have $2a\int_{0}^{\infty} e^{-|x|} = 1$.

Since $e^{-|x|} = e^{-x}$ in $(0, \infty)$, $2a\int_{0}^{\infty} e^{-x} = 1$. This implies, $a = \frac{1}{2}$

(ii) Mean = $\int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{\infty} x\frac{1}{2}e^{-|x|}dx = \frac{1}{2}\int_{-\infty}^{\infty} xe^{-|x|}dx = 0$

{Since, x $e^{-|x|}$ is an odd function}

Variance = $\int_{-\infty}^{\infty}(x - mean)^2 f(x)dx = \int_{-\infty}^{\infty} (x-0)^2 \frac{1}{2}e^{-|x|}dx = \frac{1}{2}\int_{-\infty}^{\infty} x^2e^{-|x|}dx$

$= 2\frac{1}{2}\int_{0}^{\infty} x^2e^{-|x|}dx$

{Since, $x^2e^{-|x|}$ is an even function}

$= \int_{0}^{\infty} x^2e^{-x}dx = 2$

Example 11: Find the value of 'K' and hence find mean and variance of the distribution $dF = Kx^2e^{-x}dx, \quad 0 < x < \infty$

Solution: Given $\frac{dF}{dx} = Kx^2 e^{-x}$

That is, $f(x) = Kx^2 e^{-x}$

We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx = 1$.

Now, $\int_0^{\infty} Kx^2 e^{-x} dx$ = 1.This implies, $K = \frac{1}{2}$

Mean = $\int_0^{\infty} xf(x)dx$ = $\int_0^{\infty} x\frac{1}{2}x^2 e^{-x} dx$ = 3

Variance = $\int_0^{\infty} (x - mean)^2 f(x)dx$ = $\int_0^{\infty} (x-3)^2 \frac{1}{2}x^2 e^{-x} dx$ = 3

Example 12: Verify whether the following is a distribution function.

$$F(x) = \begin{cases} 0 & ; \quad x < -a \\ \frac{1}{2}\left(\frac{x}{a}+1\right); & -a < x < a \\ 1 & ; \quad x > 1 \end{cases}$$

Solution: We know that $f(x) = \frac{dF(x)}{dx} = \begin{cases} \frac{1}{2a} & ,-a < x < a \\ 0 & ,otherwise \end{cases}$

Also, $\int_{-a}^{a} f(x)dx = \int_{-a}^{a} \frac{1}{2a} dx$ = $\frac{1}{2a}[x]_{-a}^{a}$ = $\frac{1}{2a}(2a) = 1$

Therefore, $f(x)$ is a p.d.f. and $F(x)$ is a distribution function.

Example 13: Suppose that the error in the reaction temperature in degree Celsius for a controlled laboratory experiment is a continuous random variable X having the function

$$f(x) = \begin{cases} \frac{x^2}{3} & ;\text{for} \quad -1 < x < 2 \\ 0 & ;\text{for} \quad \text{otherwise} \end{cases}$$

Obtain the cumulative distribution function $F(x)$.

Solution: We know that, cumulative distribution function, $F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)dx$

Now, $F(x) = \int_{-1}^{x} \frac{x^2}{3} dx$ = $\left[\frac{x^3}{9}\right]_{-1}^{x}$ = $\frac{x^3+1}{9}$

Therefore, cumulative distribution $F(x)=\begin{cases} 0 & \text{when} \quad x<-1 \\ \dfrac{x^3+1}{9} & \text{when} \quad -1<x<2 \\ 1 & \text{when} \quad x>2 \end{cases}$

Example 14: Suppose that the amount of money (in rupees) that a person has saved is found to be a random phenomenon with a probability function specified by the distribution function $F(x)$ as

$$F(x)=\begin{cases} 0 & \text{for} \quad x\le 1 \\ k(x-1)^4 & \text{for} \quad 1<x\le 3 \\ 1 & \text{for} \quad x>3 \end{cases}$$

(i) Find $f(x)$ (ii) Find the value of k.

Solution: (i) By definition we know that,

$$f(x)=\frac{dF(x)}{dx}=\begin{cases} 0 & \text{for} \quad x\le 1 \\ 4k(x-1)^3 & \text{for} \quad 1<x\le 3 \\ 0 & \text{for} \quad x>3 \end{cases}$$

(ii) We know that if $f(x)$ is the probability density function, then $\int_{-\infty}^{\infty} f(x)dx=1$.

Now, $\int_{-\infty}^{1} f(x)dx+\int_{1}^{3} f(x)dx+\int_{3}^{\infty} f(x)dx=1$

This implies, $0+\int_{1}^{3} 4k(x-1)^3+0=1$

This implies, $k=\dfrac{1}{16}$.

Exercise 1.2

1. Examine whether $f(x)=\dfrac{1}{\pi}\left(\dfrac{1}{1+x^2}\right); -\infty<x<\infty$ can be a probability density function of a continuous random variable.

Solution: $f(x)$ is a probability density function

2. A continuous random variable X has the following probability density function

$$f(x)=\begin{cases} 3x^2; & 0<x<1 \\ 0 \;; & \text{otherwise} \end{cases}$$

Verify that it is a p.d.f. and evaluate (i) $P(X\le \frac{1}{3})$ (ii) $P(\frac{1}{3}\le X\le \frac{1}{2})$

Solution: $f(x)$ is a p.d.f function.

$$\text{(i) } P(X \le \frac{1}{3}) = \frac{1}{27} \quad \text{(ii) } P(\frac{1}{3} \le X \le \frac{1}{2}) = \frac{19}{216}$$

3. Let T be the life time in years of new bus engines. Suppose that T is continuous with probability density function $f_T(x) = \begin{cases} 0 & ; \quad x < 1 \\ \dfrac{d}{x^3} & ; \quad x > 1 \end{cases}$

for some constant d. (a) Find the value of d (b) Find the mean of T.

Solution: a) $d = 2$ b) mean = 2

4. Find the mean of X when X is defined as $f(x) = \begin{cases} 0 & ; \quad x < 0 \\ x & ; \quad 0 \le x < 1 \\ \dfrac{3-x}{4} & ; \quad 1 \le x < 3 \\ 0 & ; \quad otherwise \end{cases}$.

Solution: mean $= \dfrac{7}{6}$

5. A continuous random variable X has the distribution function

$$F(x) = \begin{cases} 0 & ; \quad x < 0 \\ Kx^2 & ; \quad 0 \le x \le 10 \\ 100K & ; \quad x > 10 \end{cases}$$

Find (i) the constant K (ii) the probability density function $f(x)$.

Solution: (i) $K = \dfrac{1}{100}$; (ii) $f(x) = \dfrac{dF(x)}{dx} = \begin{cases} 0 & ; \quad x < 0 \\ 0.02x & ; \quad 0 \le x \le 10 \\ 0 & ; \quad x > 10 \end{cases}$

6. A random variable X has the following probability density function

$$f(x) = \begin{cases} Cx(1-x); & 0 \le x \le 1 \\ 0 \quad ; & \text{otherwise} \end{cases}$$

Find i) the constant C ii) $P(\frac{1}{2} \le X \le \frac{3}{4})$ iii) c.d.f iv) mean and variance

Solution: (i) $\mathrm{C} = 6$, (ii) $P(\frac{1}{2} \le X \le \frac{3}{4}) = \frac{11}{32}$,

(iii) $F(x) = 0$, when $x < 0$; $F(x) = 3x^2 - 2x^3$, when $0 \le x < 1$; $F(x) = 1$ when $x \ge 1$.

(iv) mean = 0.5 and variance $= 0.05$.

7. The time until a chemical reaction is complete (in milliseconds) is approximated by the cumulative distribution function $F(x) = \begin{cases} 0 & ; \quad x < 0 \\ 1 - e^{-0.01x} & ; \quad 0 \le x \end{cases}$

Determine the probability density function of X. What probability of reactions is complete within 200 milliseconds.

Solution: $f(x) = \dfrac{dF(x)}{dx} = \begin{cases} 0 & ; \quad x < 0 \\ 0.01e^{-0.01x}; & 0 \le x \end{cases}$

$P(X < 200) = F(200) = 1 - e^{-2} = 0.8647$.

8. If $f(x) = \begin{cases} xe^{-\frac{x^2}{2}} & , x \ge 0 \\ 0 & , x < 0 \end{cases}$ then show that $f(x)$ is a p.d.f and find its c.d.f $F(x)$.

Solution: $f(x)$ is a p.d.f function, $F(x) = 1 - e^{\frac{x^2}{2}}$

9. The probability distribution function of a random variable X is

$$f(x) = \begin{cases} x & ,0 < x < 1 \\ 2 - x & ,1 < x < 2 \\ 0 & ,x > 2 \end{cases}.$$

Find the cumulative distribution function of X.

Solution: $F(x) = 0$, when $x < 0$; $F(x) = \dfrac{x^2}{2}$, when $0 < x < 1$;

$F(x) = 2x - \dfrac{x^2}{2} - 1$, when $1 < x < 2$; $F(x) = 1$,when $x > 2$.

10. If the diameter of certain bolts is a continuous random variable X with the probability density function given by

$$f(x) = \begin{cases} kx & \text{when} \quad 0 < x < 2 \\ 2k & \text{when} \quad 2 < x < 4 \\ k(6 - x) & \text{when} \quad 4 < x < 6 \\ 0 & \text{when} \quad \text{otherwise} \end{cases}$$

Find the value of k and also find the cumulative distribution function $F(x)$.

Solution: $k = \dfrac{1}{8}$ and the cumulative distribution function $F(x) = 0$, when $x < 0$;

$F(x) = \dfrac{x^2}{16}$, when $0 < x < 2$; $F(x) = \dfrac{x-1}{4}$, when $2 < x < 4$;

$F(x) = \dfrac{12x - x^2 - 20}{16}$, when $1 < x < 2$; $F(x) = 1$, when $x > 6$.